



PCA Jump Model

Market Regime Detection

for RL ETF Allocation

Executive Positioning. This report reframes the project around a leak-safe PCA Jump Model regime pipeline. The regime model is the core pattern-recognition contribution; reinforcement learning is treated as the downstream allocation test that reveals both the promise and the remaining limitations of the current state representation.

Report Date	May 8, 2026
Primary Data Window	March 28, 2014 to March 13, 2026
Core Tradable Assets	SPY, TLT, GLD, with cash as an action option
Main Regime Model	PCA Jump Model with causal rolling robust scaling
Final RL Context	Existing trained artifacts only; no new training was run for this PDF

Contents

1	Final Project Report Draft	2
1.1	Executive Summary	2
1.2	Project Evolution	2
1.3	Data And Splits	3
1.4	Leakage-Safe Preprocessing	3
1.5	Why Rolling Robust Scaling Was Needed	4
1.6	PCA Compression	4
1.7	Jump Model Selection	4
1.8	Regime Interpretation	5
1.9	Smoothing And Duration	6
1.10	Mutual Information Evidence	6
1.11	RL-Ready Dataset	7
1.12	RL Setup	7
1.13	Existing RL Results	8
1.13.1	Quick Jump RL Runner	8
1.13.2	Long DQN Tuning Artifact	8
1.14	Baselines	9
1.15	Streamlit Dashboard	9
1.16	Topics That Should Not Be Missing From The Final Report	10
1.17	Limitations	11
1.18	Future Work	11
1.19	Reproducibility	11

1 Final Project Report Draft

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This report summarizes the current project state from existing artifacts only. No new RL training was run while preparing this document.

1.1 Executive Summary

This project studies whether market-regime information can improve weekly portfolio allocation across SPY, TLT, GLD, and cash. The original pipeline used an HMM regime model, but the active research path now uses a PCA Jump Model because it gives direct control over regime-switching frequency, supports an interpretable K sweep, and produces compact state features for RL and attention models.

The current leak-safe Jump Model pipeline builds weekly price and macro features, applies causal rolling robust standardization, compresses the feature set with train-fitted PCA, fits Jump Model centroids on the train split only, and assigns validation/test regimes with online causal rules. A 12-week sequence dataset is then exported for attention or RL training.

Main findings:

- The selected smoothed PCA Jump Model uses 6 PCs, $K=3$, jump penalty 32.0, and explains 53.94% of feature variance. The compression ratio is 54 original numeric market/macro features to 6 PCs, or 9.0:1.
- The selected regimes are economically interpretable: calm/risk-on, growth/trend, and stress/risk-off. The stress regime has the highest VIX and deepest SPY drawdown profile.
- $K=2$ has the best raw silhouette, while $K=5$ is the elbow choice. $K=3$ is the recommended reporting choice because it balances interpretability, duration, minimum cluster size, and attention-readiness.
- Mutual information supports using volatility, drawdown, rates, and bond-market features as regime/RL inputs, but the signal is modest and target dependent.
- Existing RL results are mixed. A quick Jump RL run performs well on validation and modestly on locked test, but the long DQN tuning artifact underperforms simple baselines. This is an important result, not just a failure: regime detection can be useful descriptively while the trading policy still needs stronger action design, reward design, and validation discipline.

1.2 Project Evolution

The proposal began with HMM-style market-regime detection and RL allocation. During experimentation, several HMM-specific assumptions became limiting:

- Gaussian emission assumptions made the states sensitive to scaling and outliers.
- Prior filtering rules constrained the model before the data structure was fully understood.
- Regime persistence was indirect, controlled through transition probabilities rather than a clear switch penalty.
- Full-sample standardization risked making early years look artificially quiet compared with later high-variance or high-volume periods.

The Jump Model path addresses these issues by clustering in PCA space while penalizing label changes through an explicit jump penalty. This makes it easier to tune the model around market-regime behavior: not only cluster separation, but also switch frequency, average duration, and usefulness for the downstream attention/RL state.

1.3 Data And Splits

The core state table is:

- `data/processed/model_state_weekly_price_macro.csv`

The train-ready Jump Model artifacts are:

- `data/processed/jump_model_train_ready_weekly.csv`
- `data/processed/jump_model_train_ready_sequences.csv`
- `data/processed/jump_model_train_ready_sequences.npz`
- `data/processed/jump_model_train_ready_metadata.json`

The split design is chronological:

split	rows	start_week	end_week
train	406	2014-03-28	2021-12-31
validation	104	2022-01-07	2023-12-29
locked test	115	2024-01-05	2026-03-13

The sequence dataset uses a 12-week lookback. A validation or test sequence may look back into prior history, which is acceptable for online inference because that history would already be known. The split assignment is based on the sequence end week, and no target values from the future are included in features.

1.4 Leakage-Safe Preprocessing

Leakage control is one of the most important report topics for this project. The active train-ready dataset avoids the major leakage paths as follows:

- Target columns such as `next_return_spy`, `next_return_tlt`, `next_return_gld`, and `best_asset_next_week` are excluded from `x_*` features.
- PCA is fit on the train split only.
- Jump Model centroids are fit on the train split only.
- Validation and locked-test regimes are assigned using current and past information only.
- Continuous RL features are standardized using train-split statistics after the causal Jump Model features are built.
- Soft-score temperature and regime naming/VIX ordering are derived from train assignments only.

The rolling raw-feature scaler is causal, not full-sample. For week t , the rolling robust scaler uses trailing observed history up to that point, with a 52-week window, at least 12 weeks of history, and clipping at ± 6.0 . This is appropriate for online inference because a deployed system can always look back at past market history. It is different from full-sample standardization, which would leak future distribution shifts into earlier observations.

1.5 Why Rolling Robust Scaling Was Needed

The first Jump Model visualizations showed many early short-lived regimes and later regimes dominated by higher variance and larger changes in volume or macro levels. This is consistent with a common clustering failure mode: nonstationary feature scale can dominate distance-based models.

The rolling robust scaler reduces this issue by standardizing each feature relative to its recent local history. Median/MAD-style scaling is less sensitive to spikes than mean/std scaling, and the clip value prevents extreme values from overpowering PCA and cluster distances. The goal is not to erase market stress; the goal is to let the model distinguish unusual moves relative to the market context available at that time.

1.6 PCA Compression

The selected Jump Model uses 6 PCA components from 54 numeric market/macro features:

item	value
original features	54
PCA components	6
compression ratio	9.0:1
explained variance	53.94%

The compression ratio is a mechanical ratio: original feature count divided by PCA component count. It should not be optimized alone. In this project, 6 PCs are preferred because they preserve more structure for downstream attention than 2 PCs, while still reducing noise and collinearity enough for stable clustering.

1.7 Jump Model Selection

The selected research Jump Model is:

parameter	value
scaler	rolling robust
scaler window	52 weeks
scaler min history	12 weeks
scaler clip	± 6.0
PCA components	6
K	3

parameter	value
jump penalty	32.0
post-cluster smoothing	3-week minimum run

Selected-K metrics:

metric	value
inertia	26732.84
silhouette	0.1844
jumps	29
min duration	3 weeks
average duration	19.30 weeks
max duration	95 weeks

The K sweep shows why this should not be chosen by silhouette alone:

K	inertia	silhouette	jumps	average duration
2	31094.35	0.3256	24	23.16
3	26732.84	0.1844	29	19.30
4	23927.78	0.1670	27	20.68
5	21622.99	0.1728	26	21.44
6	19851.10	0.1761	34	16.54
7	18346.26	0.1643	39	14.48
8	17105.95	0.1734	42	13.47
9	16622.98	0.1686	40	14.12
10	15920.72	0.1666	43	13.16

K=2 wins raw silhouette because it separates the sample into broad low-stress and high-stress areas. K=5 is the elbow-selected value. K=3 is more useful for attention/RL because it keeps a distinct stress state while avoiding too many thin regimes.

1.8 Regime Interpretation

Regime IDs are ordered from lowest average VIX to highest average VIX.

regime	interpretation	weeks	share	VIX	SPY 20d return	next SPY annualized
R0	Calm / risk-on	309	53.37%	15.44	1.93%	16.33%
R1	Growth / trend	172	29.71%	17.46	2.26%	21.02%
R2	Stress / risk-off	98	16.93%	25.29	-4.37%	-6.49%

This broadly aligns with real-world market behavior. The stress/risk-off regime has higher VIX, deeper SPY drawdown, higher SPY volatility, and negative forward SPY returns. The calm and growth regimes both support risk assets, but the growth state shows stronger trend behavior and higher forward SPY return.

The report should be careful not to claim perfect event detection. These are statistical states derived from price/macro features, not named macro events. The stronger claim is that the states have economically coherent profiles.

1.9 Smoothing And Duration

There are two smoothing concepts in the project:

1. Descriptive post-clustering smoothing for charts and static Jump Model artifacts. Runs shorter than the configured minimum duration are merged into the closest adjacent regime by PCA-centroid distance. This reduces 1-2 week visual noise in regime timelines.
2. Causal confirmation smoothing for RL features. A new regime must persist for the configured number of weeks before the confirmed regime switches. This creates a delay, but it does not inspect future weeks and is therefore acceptable for online inference.

This distinction should be highlighted in the final report. Static smoothing is for interpretation; causal smoothing is for train-ready RL state construction.

1.10 Mutual Information Evidence

Mutual information was used to check whether market/macro features contain signal for next-week allocation targets. The analysis used 579 rows and 54 numeric features, excluding close levels and future returns.

For the primary classification target, `best_asset_next_week`, the top features were:

rank	feature	MI	permutation p-value
1	<code>tnx_level</code>	0.0407	0.0198
2	<code>gld_vol_20d</code>	0.0338	0.0594
3	<code>dgs10_level</code>	0.0289	0.0495

rank	feature	MI	permutation p-value
4	tl_t_ret_1d	0.0281	0.0990
5	spy_drawdown_60d	0.0267	0.0495

For SPY next return, the strongest signals were `spy_drawdown_60d`, `vix_level`, `spy_intraday_range`, `umcsent_level`, and `spy_vol_20d`. For TLT, bond volatility and the yield curve were more important. For GLD, TLT returns and rate features were more informative.

Interpretation: the signal is real but modest. This supports using these features in a regime-aware state representation, but it also explains why a high-capacity RL agent can overfit if the action and reward setup is not tightly controlled.

1.11 RL-Ready Dataset

The train-ready dataset is designed for attention and RL:

- 12-week lookback sequences.
- 21 leak-safe features.
- PCA features, regime scores, centroid-distance information, regime duration, regime-change indicators, and target-only next-week returns.
- Targets: `y_next_return_spy`, `y_next_return_tlt`, `y_next_return_gld`, `y_best_asset_id`, and `y_best_asset`.

The attention context is important: the model should not only see the current regime label. It should also see the trajectory into that regime, distances to alternative regimes, and whether a regime transition is newly forming or already persistent.

1.12 RL Setup

The Jump RL environment uses weekly allocation actions over:

action	allocation
<code>cash_only</code>	100% cash
<code>spy_only</code>	100% SPY
<code>tlt_only</code>	100% TLT
<code>gld_only</code>	100% GLD
<code>spy_80_tlt_20</code>	80% SPY / 20% TLT
<code>balanced_60_30_10</code>	60% SPY / 30% TLT / 10% GLD
<code>defensive_20_60_20</code>	20% SPY / 60% TLT / 20% GLD

The evaluation protocol selects models using validation metrics only. Locked test is reported after selection and must not be used to choose hyperparameters.

1.13 Existing RL Results

Two relevant RL result sets exist in the repository.

1.13.1 Quick Jump RL Runner

Artifact:

- `output/full_pipeline/jump_rl_training_summary.json`

Configuration:

```
python scripts/train_jump_rl.py \
  --timesteps 30000 --seed 123 --min-action-hold-weeks 8 \
  --reward-scale 100.0 --risk-penalty 0.05
```

Metrics are net of transaction costs:

split	cumulative return	annualized return	volatility	Sharpe	max drawdown	turnover
validation	19.56%	9.34%	15.49%	0.603	-18.58%	0.115
locked test	9.65%	4.25%	14.37%	0.296	-11.90%	0.113

This run is promising because it improves validation performance while keeping turnover low. The locked-test result is positive but much weaker, so it should be reported as preliminary rather than definitive. Because this is a limited runner artifact rather than the full long-DQN tuning sweep, it should be framed as an exploratory result.

1.13.2 Long DQN Tuning Artifact

Artifact:

- `output/jump_rl_long_dqn/best_config.json`

Best selected trial:

- algorithm: DQN
- seed: 7
- timesteps: 2,000,000
- learning rate: 0.0001
- buffer size: 250,000
- batch size: 256
- gamma: 0.99

Metrics:

split	cumulative return	annualized excess return	Sharpe	max drawdown	turnover
validation	-7.85%	-6.37%	-0.430	-23.91%	0.500
locked test	-5.97%	-6.31%	-0.438	-22.33%	0.570

This long run underperforms simple baselines and has high turnover. That result suggests the current DQN setup is not yet robust, even if the regime model itself is meaningful.

1.14 Baselines

The strongest locked-test baselines are difficult to beat:

strategy	split	cumulative return	Sharpe	max drawdown
momentum_rotation_20d	validation	16.88%	0.394	-11.64%
gld_only	validation	12.76%	0.258	-17.35%
gld_only	locked test	118.10%	1.770	-14.55%
equal_weight_spy_tlt_gld	locked test	47.14%	1.439	-8.32%
momentum_rotation_20d	locked test	60.61%	1.080	-14.86%

The 2024-2026 locked-test period strongly favors GLD, which makes static GLD and simple baseline comparisons unusually strong. The report should therefore avoid claiming that the RL model is generally superior. A better conclusion is that regime-aware state construction is promising, but policy learning remains the main bottleneck.

1.15 Streamlit Dashboard

The Streamlit dashboard supports real-time-style market replay and regime diagnostics:

- market replay with indexed SPY/TLT/GLD prices
- cluster scatter view
- PCA component visualization
- regime time-series timeline
- elbow/K sweep diagnostics
- regime summary table
- heatmap-style regime feature profiles

Research defaults reflected in the dashboard:

control	default
scaler window weeks	52

control	default
scaler minimum history weeks	12
scaler clip	6.0
jump penalty	6.0
minimum displayed regime duration	6
K sweep	2 to 10
K selection	manual
manual K	4

The dashboard is useful for research exploration, but final RL artifacts should be produced by the leak-safe pipeline scripts, not by full-sample interactive refits.

1.16 Topics That Should Not Be Missing From The Final Report

The final submitted report should explicitly include these topics:

topic	why it matters
HMM-to-Jump-Model pivot	Explains why the project changed method rather than silently switching.
Leakage controls	Prevents the most common critique of financial ML projects.
Rolling robust scaling	Addresses nonstationary variance/volume scale and the user's observed early-regime noise.
PCA compression ratio	Shows the theoretical feature compression and why 6 PCs were selected.
K selection tradeoff	Explains why raw silhouette was not the only objective.
Jump penalty and duration	Shows regimes are intended to be persistent market states, not weekly noise.
Static vs causal smoothing	Clarifies what is safe for RL and what is only descriptive.
Mutual information	Provides feature-target evidence before RL.
Baseline comparison	Keeps the RL results honest.
Locked-test discipline	Shows test was not used to choose model settings.
Failure analysis	Turns weak long-DQN performance into a research insight.
Streamlit dashboard	Demonstrates inspection, replay, and interpretability.
Future RL improvements	Gives a credible path beyond current results.

1.17 Limitations

- Regime labels are statistical and should not be treated as ground-truth macro states.
- Silhouette is low for $K > 2$ because market regimes overlap; low silhouette does not automatically invalidate the regimes, but it limits confidence.
- The locked-test period is unusual because GLD performs extremely well.
- The long DQN artifact underperforms baselines, suggesting overfitting, poor exploration, action-template mismatch, or reward instability.
- Transaction costs and turnover matter materially; high-frequency regime switching can destroy performance.
- The current asset universe is small. Adding more ETFs may improve allocation opportunities but also increases action complexity.

1.18 Future Work

Recommended next steps:

1. Keep the leak-safe Jump Model dataset as the canonical regime feature source.
2. Compare RL against a simple supervised policy that predicts the best next-week asset from the same leak-safe features.
3. Add regime-aware but rule-based allocation baselines, such as stress regime to cash/GLD and calm/growth regime to SPY.
4. Tune action templates, especially reducing all-in actions and adding smoother allocation choices.
5. Train attention models with explicit regime score and regime duration inputs, then inspect attention weights by market period.
6. Evaluate across rolling walk-forward windows so the conclusion is not too dependent on the 2024-2026 GLD-dominated test period.

1.19 Reproducibility

Build the train-ready Jump Model dataset:

```
python scripts/build_train_ready_dataset.py
```

Run the Streamlit dashboard:

```
streamlit run app/streamlit_jump_model.py
```

Quick Jump RL command represented by the existing summary artifact:

```
python scripts/train_jump_rl.py \  
--timesteps 30000 --seed 123 --min-action-hold-weeks 8 \  
--reward-scale 100.0 --risk-penalty 0.05
```

Long DQN tuning artifacts are already stored under:

output/jump_rl_long_dqn/

Key supporting reports:

- reports/jump_model_train_ready_dataset.md
- reports/jump_model_tuning_rolling_robust52_smoothed.md
- reports/mutual_information_results.md
- reports/jump_rl_tuning.md